nothing about the actual attributes of the particles themselves. His approach does not fall afoul of Bell's results because, as we discussed toward the end of Chapter 4, possessing definite properties forbidden by quantum uncertainty is *not* ruled out; only locality is ruled out, and Bohm's approach is not local.[9] Instead, Bohm imagined that the wavefunction of a particle is another, *separate element of reality*, one that exists *in addition to the particleitself*. It's not particles *or* waves, as in Bohr's complementarity philosophy; according to Bohm, it's particles *and* waves. Moreover, Bohm posited that a particle's wavefunction interacts with the particle itself—it "guides" or "pushes" the particle around—in a way that determines its subsequent motion. While this approach agrees fully with the successful predictions of standard quantum mechanics, Bohm found that changes to the wavefunction in one location are able to immediately push a particle at a distant location, a finding that explicitly reveals the nonlocality of his approach. In the double-slit experiment, for example, each particle goes through one slit or the other, while its wavefunction goes through both and suffers interference. Since the wavefunction guides the particle's motion, it should not be terribly surprising that the equations show the particle is likely to land where the wavefunction value is large and it is unlikely to land where it is small, explaining the data in Figure 4.4. In Bohm's approach, there is no separate stage of wavefunction collapse since, if you measure a particle's position and find it *here*, that is truly where it was a moment before the measurement took place.

A fourth approach, developed by the Italian physicists Giancarlo Ghirardi, Alberto Rimini, and Tullio Weber, makes the bold move of modifying Schrödinger's equation in a clever way that results in hardly any effect on the evolution of wavefunctions for individual particles, but has a dramatic impact on quantum evolution when applied to "big" everyday objects. The proposed modification envisions that wavefunctions are inherently unstable; even without any meddling, these researchers suggest, sooner or later every wavefunction collapses, of its own accord, to a spiked shape. For an individual particle, Ghirardi, Rimini, and Weber postulate that wavefunction collapse happens spontaneously and randomly, kicking in, on average, only once every billion years or so.[10] This is so infrequent that it entails only the slightest change to the usual quantum mechanical description of individual